Austrian BG-trap bycatch: order-level diversity by year, season and trap type

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02 November 2026

Contents

1 Data and assumptions **1**

2 Data summary: sampling coverage and variability **1**

3 Modelling approach **3**
3.1 Response variables . . . 3
3.2 Modelling subset . . . 3
3.3 Common model structure . . . 3
3.4 Model overview . . . 4

4 Models **5**
4.1 M1: order richness . . . 5
4.2 M2: order richness adjusted for catch size . . . 5
4.3 M3: Shannon diversity, exp(H) . . . 6
4.4 M4: Diptera share . . . 6

5 Results **8**
5.1 Year trends . . . 8
5.2 Trap types and season . . . 8
5.3 Interpretation . . . 11

6 Sensitivity analyses **13**
6.1 S1: generalised Poisson for richness . . . 13
6.2 S2 and S3: region effects . . . 13
6.3 S4: annual models . . . 13

7 Conclusions and caveats **14**

8 Appendix **15**

1 Data and assumptions

The input is the `bycatch_longtable` sheet: counts of 26 taxa (insect orders plus other arthropods and slugs, all retained), pooled by region, trap type, year and month. The sampling unit is therefore one region × trap type × year × month, here called a *sampled month*. There are 183 sampled months with 59,776 individuals in total.

Zero-filling assumption (to be confirmed by the data collectors). The sheet contains no zeros, so an empty cell may mean "not sampled" or "none caught". A month is treated as sampled if at least one taxon was recorded in it; empty cells within a sampled month are set to zero. Months with no records at all are treated as not sampled.

2 Data summary: sampling coverage and variability

Key points for the data collectors:

- **Catch varies by orders of magnitude within a series.** In Vienna, BG Sentinel + CO2, monthly catch ranges from 2 to 553 individuals (276-fold). Without trapping effort we cannot tell fewer insects from fewer traps or trap-nights (Table T5).
- **Recording protocols differ.** Salzburg and Styria list only 2 to 3 taxa; Upper Austria (CO2, 2014-2015) and Tyrol record 6 and 5 of 26 (Table T4).
- **Coverage is uneven.** Months sampled per year range from 1 to 7, and April was sampled only in 2014 and 2015 (Tables T1 and T2). Three months are missing inside otherwise sampled seasons (Table T3).
- **Trap type, region and period overlap.** Each trap type was used in different regions and years (Table T7), so a trap effect cannot be fully separated from a place or period effect.
- **The catch is dominated by Diptera** (78% of all individuals), and 15 of the 26 taxa total 30 individuals or fewer.

T1. Number of months sampled per year.

region	trap	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Burgenland	BG Sentinel + CO2	6	7	5	5	2	6	6	6	6	5	6	0
Lower Austria	BG Sentinel + CO2	6	6	0	0	0	0	0	0	0	0	0	0
Lower Austria	BG Sentinel + Yeast	0	0	0	0	0	0	0	4	4	3	0	4
Salzburg	BG Sentinel + CO2	0	0	0	0	0	4	0	0	0	0	0	0
Styria	BG Sentinel + CO2	0	0	0	0	0	4	0	0	0	0	0	0
Tyrol	BG Pro + Yeast	0	0	0	0	0	0	0	0	0	0	0	3
Upper Austria	BG Sentinel + CO2	2	1	0	0	0	0	0	0	0	0	0	0
Upper Austria	BG Sentinel + CO2 + Lure	0	0	0	0	0	0	5	5	5	5	0	0
Vienna	BG Sentinel + CO2	6	7	0	0	0	4	0	0	0	0	0	0
Vienna	BG Sentinel + CO2 + Lure	0	0	0	4	4	5	5	5	4	5	2	5
Vienna	BG Sentinel + Yeast	0	0	0	0	0	0	0	2	4	0	0	0

T2. Number of sampled months by calendar month, pooled over years.

region	trap	Apr	May	Jun	Jul	Aug	Sep	Oct
Burgenland	BG Sentinel + CO2	2	11	11	10	9	9	8
Lower Austria	BG Sentinel + CO2	1	2	2	2	2	2	1
Lower Austria	BG Sentinel + Yeast	0	1	1	4	4	3	2
Salzburg	BG Sentinel + CO2	0	0	1	1	1	1	0
Styria	BG Sentinel + CO2	0	0	1	1	1	1	0
Tyrol	BG Pro + Yeast	0	0	0	0	1	1	1
Upper Austria	BG Sentinel + CO2	0	1	1	1	0	0	0
Upper Austria	BG Sentinel + CO2 + Lure	0	4	4	4	4	4	0
Vienna	BG Sentinel + CO2	1	2	3	3	3	3	2
Vienna	BG Sentinel + CO2 + Lure	0	8	8	8	8	7	0
Vienna	BG Sentinel + Yeast	0	0	0	1	2	2	1

T3. Unsampled months inside an otherwise sampled season: true zero catch or no trapping?

region	trap	year	missing_months
Burgenland	BG Sentinel + CO2	2016	Aug, Sep
Lower Austria	BG Sentinel + Yeast	2025	Jun
Vienna	BG Sentinel + CO2 + Lure	2022	Jun

T4. Taxa listed in the data template and taxa recorded at least once.

region	trap	taxa_listed	taxa_recorded
Burgenland	BG Sentinel + CO2	26	22
Lower Austria	BG Sentinel + CO2	26	20
Lower Austria	BG Sentinel + Yeast	26	15
Salzburg	BG Sentinel + CO2	2	2
Styria	BG Sentinel + CO2	3	3
Tyrol	BG Pro + Yeast	26	5
Upper Austria	BG Sentinel + CO2	26	6
Upper Austria	BG Sentinel + CO2 + Lure	26	16
Vienna	BG Sentinel + CO2	26	15
Vienna	BG Sentinel + CO2 + Lure	26	16
Vienna	BG Sentinel + Yeast	26	10

T5. Catch size (individuals per sampled month), richness and Diptera share by region and trap type.

region	trap	months	years	N_median	N_min	N_max	max_min_ratio	N_CV	richness_median	richness_range	diptera_share
Burgenland	BG Sentinel + CO2	60	2014-2024	351.5	8	1266	158.2	0.68	8.0	1-13	0.76
Lower Austria	BG Sentinel + CO2	12	2014-2015	818.0	33	1322	40.1	0.54	10.0	6-12	0.84
Lower Austria	BG Sentinel + Yeast	15	2021-2025	196.0	7	842	120.3	0.87	8.0	2-11	0.74
Salzburg	BG Sentinel + CO2	4	2019	9.0	3	18	6.0	0.72	1.0	1-2	1.00
Styria	BG Sentinel + CO2	4	2019	7.0	4	14	3.5	0.54	1.5	1-2	0.96
Tyrol	BG Pro + Yeast	3	2025	40.0	23	156	6.8	0.99	3.0	2-4	0.91
Upper Austria	BG Sentinel + CO2	3	2014-2015	30.0	8	38	4.8	0.61	3.0	2-5	0.80
Upper Austria	BG Sentinel + CO2 + Lure	20	2020-2023	159.0	81	770	9.5	0.75	8.0	5-11	0.76
Vienna	BG Sentinel + CO2	17	2014-2019	80.0	2	553	276.5	1.06	4.0	1-11	0.86
Vienna	BG Sentinel + CO2 + Lure	39	2017-2025	280.0	48	975	20.3	0.72	8.0	3-12	0.82
Vienna	BG Sentinel + Yeast	6	2021-2022	149.5	87	221	2.5	0.37	6.5	3-9	0.83

T7a. Sampled months per trap type and year.

trap	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
BG Pro + Yeast	0	0	0	0	0	0	0	0	0	0	0	3
BG Sentinel + CO2	20	21	5	5	2	18	6	6	6	5	6	0
BG Sentinel + CO2 + Lure	0	0	0	4	4	5	10	10	9	10	2	5
BG Sentinel + Yeast	0	0	0	0	0	0	0	6	8	3	0	4

T7b. Sampled months per trap type and region.

trap	Tyrol	Burgenland	Lower Austria	Salzburg	Styria	Upper Austria	Vienna
BG Pro + Yeast	3	0	0	0	0	0	0
BG Sentinel + CO2	0	60	12	4	4	3	17
BG Sentinel + CO2 + Lure	0	0	0	0	0	20	39
BG Sentinel + Yeast	0	0	15	0	0	0	6

Table T6 (catch per year) is in the Appendix.

3 Modelling approach

3.1 Response variables

Three diversity responses are computed for each sampled month:

- **Order richness** (S): number of taxa with at least one individual. Sensitive to rare taxa, and therefore to sampling effort.
- **Shannon diversity**, reported as **exp(H)**: the *effective number of orders*, i.e. the number of equally common orders that would give the observed Shannon index H. Unlike S, it weights orders by their abundance.
- **Diptera share**: the proportion of individuals that are Diptera.

These responses are not independent. Richness rises strongly with catch size (r = 0.73 with log catch), whereas Shannon H barely does (r = 0.1). Shannon is driven instead by the Diptera share (r = -0.74).

3.2 Modelling subset

Salzburg and Styria are excluded because their richness is capped by the recording protocol (Table T4). BG Pro + Yeast is excluded because it was used only in Tyrol in 2025 (3 months), so no trend or season can be estimated for it. The models use **172 sampled months** from 4 regions (Burgenland, Lower Austria, Upper Austria, Vienna) and 3 trap types.

3.3 Common model structure

All monthly models share the same fixed-effect structure:

`response ~ trap × year + trap × (month + month²) + (1 | region)`

Each term answers a specific question:

Model terms and the question each one tests.

term	question
trap	Do trap types differ, in July 2022?
year	Is there a trend across years (for the CO2 trap)?
trap × year	Does the year trend differ between trap types?
month, month²	Is there a seasonal pattern, with a single peak or trough?
trap × month, trap × month²	Does the seasonal pattern differ between trap types?
(1 | region)	Regions differ in their baseline level; not a question, a nuisance term.

Year and month are expressed relative to a reference point, 2022 and July (year - 2022, month - 7; both remain whole numbers). This does not change trends, predictions or the tests of interactions. It only means that the trap differences in the tables refer to July 2022, a point inside every trap's series, rather than to "year 0". All results below are reported in natural units.

Assumptions shared by all monthly models:

1. Empty cells within a sampled month are true zeros (Section 1).
2. **Sampling effort is constant** across regions, traps and years. This is the strongest assumption and cannot be checked without effort data.
3. The trend is linear on the model scale (constant % change per year), and the season is a smooth curve with at most one peak.
4. Sampled months are independent once region is accounted for. This is checked in Section 6.
5. With only 4 regions, the region variance is poorly estimated. This is checked in Section 6.

3.4 Model overview

Overview of the monthly models. All share the structure trap × year + trap × (month + month²) + (1 | region).

model	question	response	distribution	extra_terms	n
M1	Does the number of orders change with year and season, and differently by trap type?	S (count)	Poisson, log link		172
M2	Does it change once differences in catch size are accounted for?	S (count)	Poisson, log link	+ log catch	172
M3	Does the effective number of orders, exp(H), change with year and season?	H (exp(H) reported)	Gaussian, identity link on H		172
M4	Does the share of Diptera in the catch change with year and season?	Diptera / total individuals	Beta-binomial, logit link		172

4 Models

4.1 M1: order richness

- **Question.** Does the number of orders caught per month change across years and seasons, and does this differ between trap types?
- **Response.** Number of taxa with at least one individual (count, 0 to 26).
- **Distribution.** Poisson with log link: richness is a count, and coefficients become % changes.
- **Dispersion.** The Poisson assumes variance equal to the mean. Richness is a bounded count and is *under*-dispersed here (ratio below). We keep the Poisson and do not correct for this: under-dispersion makes Poisson standard errors too wide, so tests are conservative. The effect on the conclusions is checked in Section 6. A negative binomial is not an option because it can only model over-dispersion; fitted here, it collapses to the Poisson.
- **Specific assumptions.** Richness depends on how many individuals were caught, which this model ignores (see M2).

Pearson dispersion ratio: **0.6** (1 = Poisson-like; < 1 = under-dispersed).

M1. Order richness, Poisson GLMM (log scale). Lure and Yeast: difference from CO2 in July 2022; × year and × month terms: difference in slope from CO2.

term	estimate	SE	z	p
intercept (CO2, July 2022)	1.968	0.153	12.890	<1e-04
Lure	0.357	0.154	2.324	0.0201
Yeast	-0.087	0.173	-0.502	0.6155
year	0.003	0.013	0.213	0.8309
month	-0.057	0.023	-2.522	0.0117
month²	-0.015	0.013	-1.138	0.2550
Lure × year	-0.017	0.025	-0.688	0.4914
Yeast × year	-0.006	0.063	-0.089	0.9294
Lure × month	0.081	0.041	1.997	0.0458
Yeast × month	0.348	0.129	2.685	0.0072
Lure × month²	-0.013	0.031	-0.409	0.6822
Yeast × month²	-0.141	0.053	-2.654	0.0079

4.2 M2: order richness adjusted for catch size

- **Question.** Are differences in richness explained simply by how many individuals were caught? Catch size is used as a proxy for the unknown sampling effort.
- **Response and distribution.** As in M1 (Poisson, log link, no correction for under-dispersion).
- **Extra term.** log(total catch).
- **Specific assumptions.** Catch size reflects both effort and true insect abundance. Adjusting for it removes effort differences, but it may also remove genuine abundance-driven changes in diversity. M2 is therefore conservative: an effect that survives M2 is unlikely to be an effort artefact.

Pearson dispersion ratio: **0.41**.

M2. Order richness adjusted for log catch, Poisson GLMM (log scale). Terms as in M1.

term	estimate	SE	z	p
intercept (CO2, July 2022)	0.803	0.184	4.364	<1e-04
Lure	-0.042	0.101	-0.411	0.68
Yeast	0.030	0.143	0.207	0.84
year	0.003	0.011	0.301	0.76
month	-0.030	0.023	-1.280	0.20
month^2	-0.017	0.013	-1.252	0.21
log catch	0.230	0.029	7.822	<1e-04
Lure × year	-0.006	0.025	-0.242	0.81
Yeast × year	0.007	0.063	0.105	0.92
Lure × month	0.059	0.041	1.460	0.14
Yeast × month	0.178	0.136	1.301	0.19
Lure × month^2	0.031	0.031	0.991	0.32
Yeast × month^2	-0.077	0.056	-1.372	0.17

4.3 M3: Shannon diversity, exp(H)

- **Question.** Does the effective number of orders, $\exp(H)$, change across years and seasons, and does this differ between trap types?
- **Response.** Shannon index H per sampled month.
- **Distribution.** Gaussian with identity link on H. Because $H = \log(\exp(H))$, this is a log-scale model of $\exp(H)$: a coefficient β means $\exp(H)$ is multiplied by $\exp(\beta)$, so results are reported as % changes in the effective number of orders, as for richness.
- **Specific assumptions.** Residuals of H are approximately normal with constant variance. H is bounded at zero (months with a single order), which the Gaussian ignores; only 2 sampled months have $H = 0$. Shannon depends much less on catch size than richness does, so no catch-size adjustment is made.

M3. Shannon H, Gaussian GLMM. exp(estimate) = multiplicative change in exp(H). Terms as in M1.

term	estimate	SE	z	p
intercept (CO2, July 2022)	0.829	0.092	8.993	<1e-04
Lure	0.034	0.119	0.288	0.773
Yeast	0.188	0.130	1.446	0.148
year	0.010	0.011	0.968	0.333
month	-0.036	0.018	-2.058	0.040
month^2	-0.015	0.010	-1.438	0.150
Lure × year	-0.027	0.020	-1.347	0.178
Yeast × year	-0.068	0.051	-1.333	0.182
Lure × month	0.008	0.032	0.254	0.800
Yeast × month	0.070	0.077	0.906	0.365
Lure × month^2	0.032	0.025	1.311	0.190
Yeast × month^2	-0.060	0.032	-1.873	0.061

4.4 M4: Diptera share

- **Question.** Does the proportion of Diptera in the catch change across years and seasons, and does this differ between trap types? Since Diptera make up most of the catch, this asks whether the balance between Diptera and all other taxa is stable.
- **Response.** Number of Diptera out of the total number of individuals in each sampled month.
- **Distribution.** Beta-binomial with logit link. A binomial model would assume that each individual is an independent draw. Insects arrive in clumps (swarms, emergences, single nights with large catches), so the variation between months is far larger than binomial: the binomial dispersion ratio is computed below. The beta-binomial adds a parameter for this extra variation. Unlike richness, this over-dispersion is too large to ignore.
- **Specific assumptions.** The log-odds of being a Diptera change linearly with year and smoothly with season. Every individual counts equally, so months with large catches carry more weight.
- **Interpretation of coefficients.** On the log-odds scale; results are reported as predicted percentages.

Binomial Pearson dispersion ratio: **23.7**, hence the beta-binomial.

M4. Diptera share, beta-binomial GLMM (log-odds scale). Terms as in M1.

term	estimate	SE	z	p
intercept (CO2, July 2022)	0.989	0.169	5.854	<1e-04
Lure	0.356	0.260	1.369	0.171
Yeast	-0.290	0.275	-1.058	0.290
year	-0.039	0.024	-1.624	0.104
month	0.067	0.040	1.672	0.094
month²	0.029	0.024	1.198	0.231
Lure × year	0.067	0.044	1.516	0.130
Yeast × year	0.138	0.113	1.221	0.222
Lure × month	-0.001	0.071	-0.017	0.986
Yeast × month	0.072	0.202	0.356	0.722
Lure × month²	-0.086	0.055	-1.567	0.117
Yeast × month²	0.067	0.089	0.750	0.454

5 Results

5.1 Year trends

R1. Change per year in order richness and in the effective number of orders, exp(H), with 95% confidence intervals.

response	trap	% per year	lower	upper
M1 richness	BG Sentinel + CO2	0.3	-2.3	3.0
	BG Sentinel + CO2 + Lure	-1.4	-5.5	2.8
	BG Sentinel + Yeast	-0.3	-11.5	12.4
M2 richness | catch	BG Sentinel + CO2	0.3	-1.8	2.6
	BG Sentinel + CO2 + Lure	-0.3	-4.5	4.1
	BG Sentinel + Yeast	1.0	-10.6	14.2
M3 exp(H)	BG Sentinel + CO2	1.0	-1.1	3.2
	BG Sentinel + CO2 + Lure	-1.7	-5.0	1.7
	BG Sentinel + Yeast	-5.6	-14.5	4.2

R2. Predicted values in July at the first and last year of each trap's series (M1, M3, M4), with 95% confidence intervals.

trap	year	orders	exp(H)	Diptera
BG Sentinel + CO2	2014	7.0 (5.3-9.2)	2.11 (1.82-2.44)	79% (74-83)
BG Sentinel + CO2	2024	7.2 (5.2-10.0)	2.34 (1.89-2.89)	71% (62-79)
BG Sentinel + CO2 + Lure	2017	11.0 (8.0-15.2)	2.58 (2.08-3.20)	77% (69-84)
BG Sentinel + CO2 + Lure	2025	9.8 (7.0-13.7)	2.25 (1.81-2.80)	81% (73-87)
BG Sentinel + Yeast	2021	6.6 (4.4-9.8)	2.93 (2.24-3.82)	65% (51-76)
BG Sentinel + Yeast	2025	6.5 (4.2-10.1)	2.33 (1.71-3.17)	73% (59-84)

5.2 Trap types and season

R3. Predicted values per trap type in July 2022, with 95% confidence intervals. The last column (M2) assumes the same catch size, 265 individuals, for every trap.

trap	orders	exp(H)	Diptera	orders at median catch
BG Sentinel + CO2	7.2 (5.3-9.7)	2.29 (1.91-2.75)	73% (66-79)	8.1 (7.0-9.2)
BG Sentinel + CO2 + Lure	10.2 (7.7-13.6)	2.37 (2.00-2.81)	79% (74-84)	7.7 (6.7-9.0)
BG Sentinel + Yeast	6.6 (4.6-9.3)	2.76 (2.22-3.44)	67% (57-75)	8.3 (6.5-10.6)

R4. Type III Wald chi-square tests for each model term.

response	term	Chisq	Df	p
M1 richness	trap	8.10	2	0.0174
	year	0.05	1	0.8309
	month	6.36	1	0.0117
	month²	1.30	1	0.2550
	trap × year	0.47	2	0.7892
	trap × month	10.25	2	0.0059
	trap × month²	7.06	2	0.0293
M2 richness | catch	trap	0.30	2	0.8611
	year	0.09	1	0.7633
	month	1.64	1	0.2004
	month²	1.57	1	0.2107
	log catch	61.18	1	<1e-04
	trap × year	0.07	2	0.9637
	trap × month	3.48	2	0.1758
	trap × month²	3.25	2	0.1969
M3 Shannon	trap	2.32	2	0.3138
	year	0.94	1	0.3330
	month	4.24	1	0.0396
	month²	2.07	1	0.1504
	trap × year	3.26	2	0.1960
	trap × month	0.84	2	0.6566
	trap × month²	6.00	2	0.0499
M4 Diptera share	trap	6.23	2	0.0443
	year	2.64	1	0.1043
	month	2.80	1	0.0945
	month²	1.44	1	0.2309
	trap × year	3.50	2	0.1736
	trap × month	0.13	2	0.9370
	trap × month²	3.36	2	0.1863

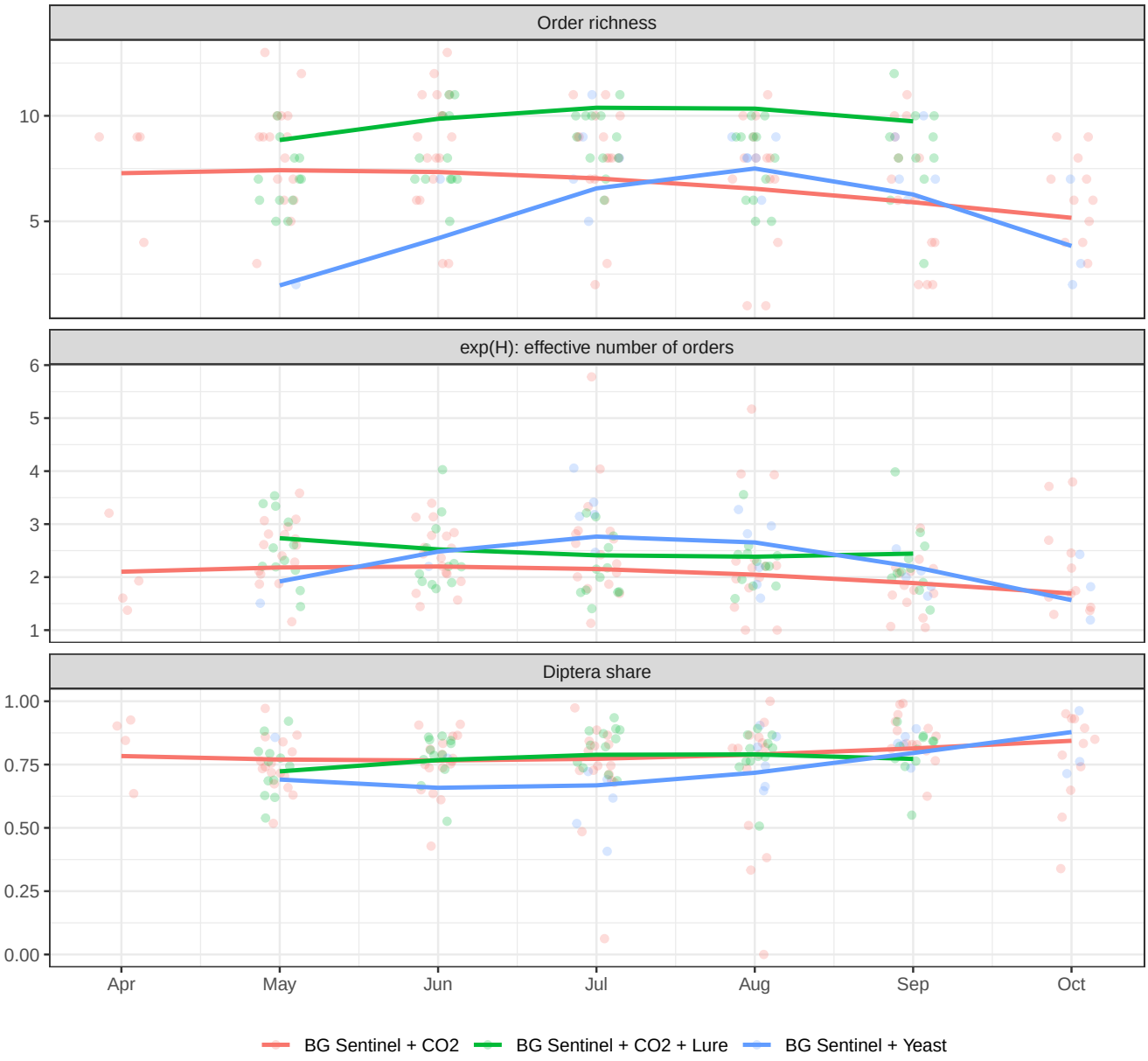


Figure 1: Predicted values (region effect set to zero) by month, at the median year of each trap's series, with observed sampled months as points.

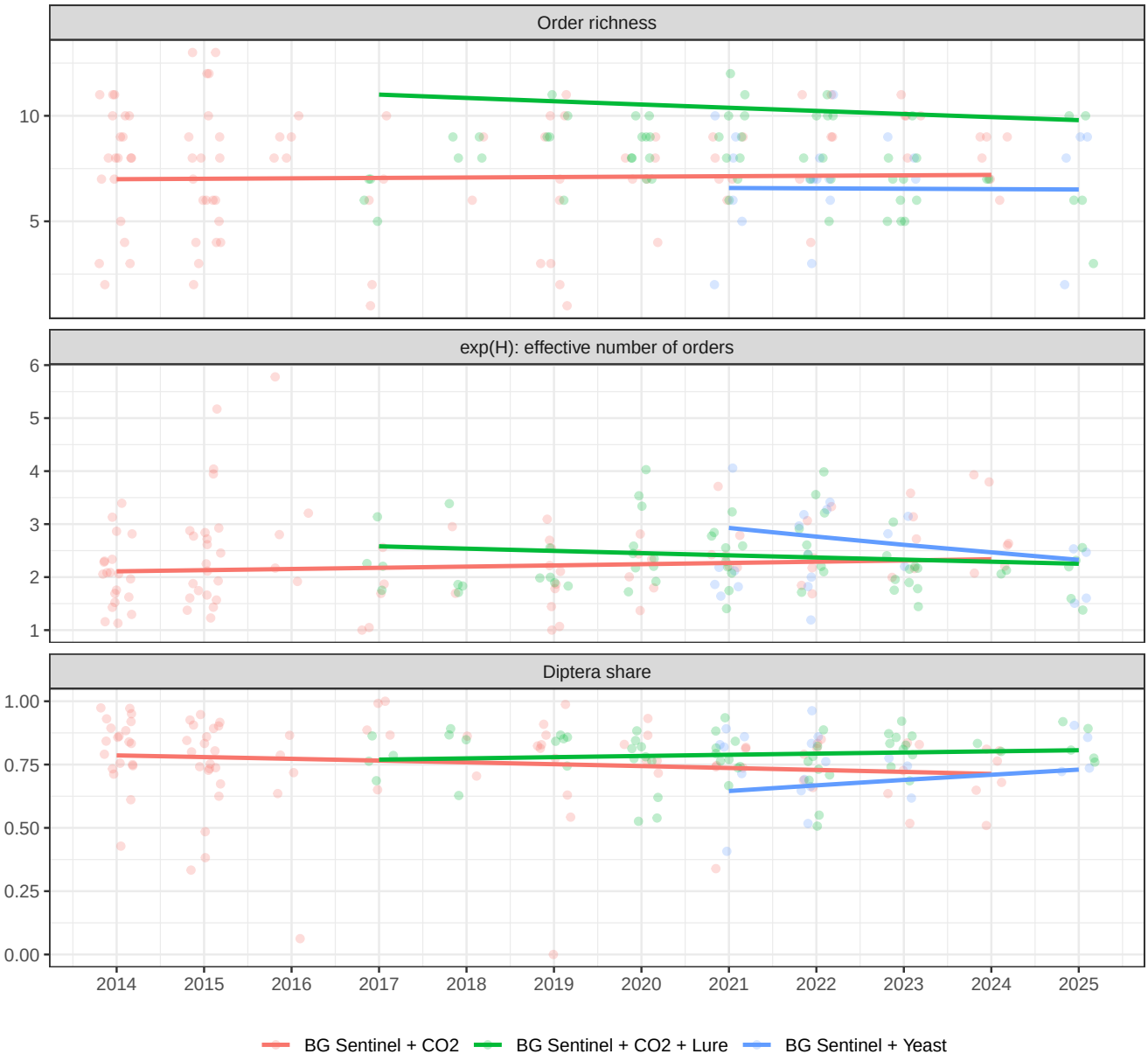


Figure 2: Predicted values for July across the years covered by each trap type, with observed sampled months (all calendar months) as points.

5.3 Interpretation

- **No temporal trend.** No response changes with year for any trap type (trap × year: M1 p = 0.79, M3 p = 0.2, M4 p = 0.17). For the CO2 trap, the longest series, richness changes by +0.3% per year (95% CI -2.3 to +3.0) and exp(H) by +1.0% per year (95% CI -1.1 to +3.2) (Table R1). In July, predicted richness for this trap is 7.0 (5.3-9.2) orders in 2014 and 7.2 (5.2-10.0) in 2024 (Table R2).
- **The Yeast trend cannot be estimated.** Its series covers 4 years; the interval for richness, -0.3% per year (95% CI -11.5 to +12.4), is too wide to support "no trend".
- **The seasonal pattern of richness differs between trap types** in M1 (trap × month p = 0.0059; trap × month² p = 0.029). Predicted richness is highest in May for CO2, July for CO2 + Lure and August for Yeast. Once catch size is accounted for (M2), this difference is not significant (smallest p = 0.18). The seasonal differences between traps are therefore largely differences in how many individuals each trap catches through the season.

- **The Lure's higher richness comes from its larger catch.** In July 2022, CO2 + Lure is predicted to catch 10.2 orders against 7.2 for CO2 alone (M1, trap p = 0.017). At the same catch size the difference shrinks to 7.7 vs 8.1 (M2, trap p = 0.86). The Lure attracts more individuals, not a more diverse mix of them.
- **Shannon diversity, exp(H), is stable.** It does not differ between trap types (p = 0.31) and has at most a weak seasonal signal (month p = 0.04; trap × month² p = 0.05).
- **The Diptera share does not change over time or season** (year p = 0.1; month p = 0.094; trap × year p = 0.17). Trap types differ slightly in level (p = 0.044): in July 2022 the predicted share is 73% (66-79) for CO2, 79% (74-84) for CO2 + Lure and 67% (57-75) for Yeast (Table R3). Since Shannon largely tracks the Diptera share (r = -0.74), M3 and M4 tell a similar story.

6 Sensitivity analyses

6.1 S1: generalised Poisson for richness

- **Question.** Does ignoring the under-dispersion of richness change the conclusions of M1 and M2?
- **Approach.** Refit M1 and M2 with a generalised Poisson, which estimates the dispersion instead of fixing it at 1.

S1. p-values of each term under the Poisson (main models) and the generalised Poisson.

response	term	p_Poisson	p_genpois
M1 richness	trap	0.0174	0.00031
	year	0.8309	0.92723
	month	0.0117	0.00095
	month²	0.2550	0.09251
	trap × year	0.7892	0.78060
	trap × month	0.0059	0.00012
	trap × month²	0.0293	0.00291
M2 richness | catch	trap	0.8611	0.65677
	year	0.7633	0.89436
	month	0.2004	0.05135
	month²	0.2107	0.09345
	log catch	<1e-04	< 1e-04
	trap × year	0.9637	0.96344
	trap × month	0.1758	0.00387
	trap × month²	0.1969	0.00997

- **Year trends do not change.** With the generalised Poisson, the CO2 richness trend is +0.1% per year (95% CI -1.8 to +2.0): same conclusion, narrower interval.
- **The seasonal difference between traps in M2 depends on the distribution.** With the generalised Poisson, trap × month is p = 0.00387 and trap × month² p = 0.00997 in M2, against 0.1758 and 0.1969 with the Poisson. This result should be treated as uncertain.

6.2 S2 and S3: region effects

- **S2. Question:** does treating region as random matter with only 4 regions? Refitting M1 with region as a fixed effect changes coefficients by at most 0.16 on the log scale (Table A1).
- **S3. Question:** are months within the same region and year more alike than the model assumes? Adding a region × year random intercept to M1 gives AIC 805 against 803 without it, so it adds nothing and the independence assumption is acceptable.

6.3 S4: annual models

- **Question.** Does the "no trend" result depend on the seasonal term, or on the uneven number of months sampled per year?
- **Approach.** Counts are summed over a fixed window (July to September) for years in which all three months were sampled; the season term is dropped. June to August would leave the Yeast trap with a single year.
- **Responses and distributions.** Richness adjusted for log catch (Poisson, as M2); richness rarefied to a common number of individuals (Gaussian); Shannon H (Gaussian, as M3); Diptera share (beta-binomial, as M4).
- **Assumptions.** A three-month total is comparable across years. Rarefaction assumes that individuals are a random sample from the trap's catch; years below the rarefaction depth are dropped.

Years available: BG Sentinel + CO2 14, BG Sentinel + CO2 + Lure 11, BG Sentinel + Yeast 4; 3 years fall below 300 individuals and are dropped from the rarefied model.

S4. Annual models (July to September): change per year with 95% confidence intervals.

response	unit	trap	change	lower	upper
Richness | log catch	% per year	BG Sentinel + CO2	-1.02	-5.67	3.85
		BG Sentinel + CO2 + Lure	0.56	-8.13	10.08
		BG Sentinel + Yeast	3.82	-14.54	26.12
Rarefied richness (300 ind.)	orders per year	BG Sentinel + CO2	-0.04	-0.18	0.09
		BG Sentinel + CO2 + Lure	-0.02	-0.25	0.22
		BG Sentinel + Yeast	-0.13	-0.67	0.41
exp(H)	% per year	BG Sentinel + CO2	0.63	-2.75	4.13
		BG Sentinel + CO2 + Lure	0.37	-5.87	7.02
		BG Sentinel + Yeast	-4.79	-17.86	10.36
Diptera share	% change in odds per year	BG Sentinel + CO2	-1.18	-8.03	6.18
		BG Sentinel + CO2 + Lure	-0.39	-12.49	13.38
		BG Sentinel + Yeast	6.75	-17.83	38.67

- **The annual models agree with the monthly ones**: no year trend differs from zero, for any response or trap type.
- With 29 rows, the region variance is close to zero in these models; they behave as ordinary GLMs.

7 Conclusions and caveats

- At order level, there is **no detectable change in diversity between 2014 and 2025** for the CO2 and CO2 + Lure traps, whether measured as richness, effective number of orders (exp(H)) or Diptera share. For the Yeast trap the series is too short to say.
- **Trap types differ in the seasonal pattern of richness**, but part or all of this reflects how many individuals each trap catches through the season (S1). The Lure's higher richness reflects its larger catch.
- **The balance between Diptera and other taxa is stable** across years and seasons. Trap types differ slightly in Diptera share, with the Yeast trap catching proportionally fewer Diptera.
- All results are **conditional on the zero-filling assumption and on constant sampling effort**. Richness depends on catch size, so effort data could change the richness results; Shannon and the Diptera share are less affected.
- **Trap type is confounded with region and period**, so trap comparisons are partly comparisons of places and years.
- **Order level is coarse.** With Diptera making up 78% of the catch, changes within Diptera (or within any order) are invisible here.
- **Environmental drivers** should wait for effort data, trap coordinates and collection dates.

8 Appendix

T6. Catch per year by region and trap type.

region	trap	year	months	N_total	N_per_month	richness_max
Burgenland	BG Sentinel + CO2	2014	6	3060	510	9
Burgenland	BG Sentinel + CO2	2015	7	1512	216	13
Burgenland	BG Sentinel + CO2	2016	5	1255	251	10
Burgenland	BG Sentinel + CO2	2017	5	1373	275	10
Burgenland	BG Sentinel + CO2	2018	2	248	124	9
Burgenland	BG Sentinel + CO2	2019	6	3116	519	11
Burgenland	BG Sentinel + CO2	2020	6	2217	370	9
Burgenland	BG Sentinel + CO2	2021	6	2054	342	9
Burgenland	BG Sentinel + CO2	2022	6	3701	617	11
Burgenland	BG Sentinel + CO2	2023	5	3103	621	11
Burgenland	BG Sentinel + CO2	2024	6	1136	189	9
Lower Austria	BG Sentinel + CO2	2014	6	5520	920	11
Lower Austria	BG Sentinel + CO2	2015	6	3444	574	12
Lower Austria	BG Sentinel + Yeast	2021	4	717	179	10
Lower Austria	BG Sentinel + Yeast	2022	4	1486	372	11
Lower Austria	BG Sentinel + Yeast	2023	3	1089	363	9
Lower Austria	BG Sentinel + Yeast	2025	4	832	208	9
Salzburg	BG Sentinel + CO2	2019	4	39	10	2
Styria	BG Sentinel + CO2	2019	4	32	8	2
Tyrol	BG Pro + Yeast	2025	3	219	73	4
Upper Austria	BG Sentinel + CO2	2014	2	46	23	3
Upper Austria	BG Sentinel + CO2	2015	1	30	30	5
Upper Austria	BG Sentinel + CO2 + Lure	2020	5	1759	352	10
Upper Austria	BG Sentinel + CO2 + Lure	2021	5	1345	269	9
Upper Austria	BG Sentinel + CO2 + Lure	2022	5	1180	236	11
Upper Austria	BG Sentinel + CO2 + Lure	2023	5	1018	204	10
Vienna	BG Sentinel + CO2	2014	6	1364	227	11
Vienna	BG Sentinel + CO2	2015	7	1554	222	9
Vienna	BG Sentinel + CO2	2019	4	121	30	3
Vienna	BG Sentinel + CO2 + Lure	2017	4	666	166	7
Vienna	BG Sentinel + CO2 + Lure	2018	4	1993	498	9
Vienna	BG Sentinel + CO2 + Lure	2019	5	2469	494	11
Vienna	BG Sentinel + CO2 + Lure	2020	5	1286	257	10
Vienna	BG Sentinel + CO2 + Lure	2021	5	3402	680	12
Vienna	BG Sentinel + CO2 + Lure	2022	4	1517	379	10
Vienna	BG Sentinel + CO2 + Lure	2023	5	1140	228	8
Vienna	BG Sentinel + CO2 + Lure	2024	2	440	220	7
Vienna	BG Sentinel + CO2 + Lure	2025	5	1409	282	10
Vienna	BG Sentinel + Yeast	2021	2	387	194	9
Vienna	BG Sentinel + Yeast	2022	4	497	124	7

A1. M1 coefficients with region as random (main model) vs fixed effect (S2).

term	random_region	fixed_region
intercept (CO2, July 2022)	1.968	2.133
Lure	0.357	0.433
Yeast	-0.087	-0.090
year	0.003	0.001
month	-0.057	-0.057
month²	-0.015	-0.016
Lure × year	-0.017	-0.015
Yeast × year	-0.006	-0.010
Lure × month	0.081	0.081
Yeast × month	0.348	0.353
Lure × month²	-0.013	-0.012
Yeast × month²	-0.141	-0.142